

Caden (ZiHuan) Zhang

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EDUCATION

University of British Columbia

Bachelor of Science in Computer Science - GPA: 4.08/4.33 (94.3%)

Vancouver, BC

Sept. 2024 – April 2028

TECHNICAL WORK EXPERIENCE

Full Stack Developer — Checker Software

January 2026 – August 2026

- Developed a web-based **DOCX renderer** with **Rust**, speeding up document comparison by **60%**
- Optimized **LibreOffice WASM** for the web, lowering bundle size by **25%**
- Developed the **localization system** for the Diffchecker desktop app, enabling multi-language support

SWE Highschool Internship — Nokia

July 2023 – August 2023

- Improved stability and performance of the Netguard Endpoint Detection and Response product by identifying and resolving 3 recurring performance bottlenecks across **Linux VMs** using **Grafana** and **Prometheus**
- Automated Jira ticket tracking with **PowerBI dashboards**, improving team workflow visibility
- Increased test coverage by 10% through implementing **RED Team assessments** with **Caldera** and **Robot Framework**, enhancing product reliability across multiple platforms

PROJECTS

Myelin Notes | React, TypeScript, Tauri, Rust

Dec. 2024 – Present

- A combination of **Goodnote** and **Obsidian** - a cross platform note-taking and knowledge management solution; shipped **13 public releases** through automated **GitHub Actions** pipelines
- Implemented on-device **semantic search** by running the **all-MiniLM-L6-v2** embedding model locally with **Candle** (Rust), backed by a background indexing engine with debouncing, bounded concurrency, and staleness hashing
- Built live **audio transcription** of streamed microphone input with **whisper.cpp**, using a quantized model, voice-activity detection, and optional **Vulkan** GPU acceleration with CPU fallback
- Developed conflict-free peer-to-peer note syncing with **Yjs CRDTs** over the **iroh** P2P networking stack

Minecraft Modding Libraries | Java, JSON, OOP, API Design, Documentation

Aug. 2021 – Dec. 2024

- Created and maintained a suite of libraries and mods that streamlined Minecraft modding workflows, improving development efficiency by **20–30%**
- Achieved a combined total of over **360,000 downloads** across all published projects
- Designed and maintained **public-facing APIs**, ensuring extensibility, stability, and developer-friendly integration

Party Lab | C++, Raylib, Lua, Steam API

July 2024 – Dec. 2024

- Developed a framework/hub for multiplayer games with declarative UI API using **C++** and **Raylib**
- Integrated **Lua** scripting via the sol2 library within a sandboxed environment, enabling secure modding and custom game modes
- Implemented multiplayer networking with Valve's **GameNetworkingSockets API** and integrated the **Steam API** for friend invites and leaderboards

Sand of Souls | Godot, GDScript, GLSL

July 2026

- Survivor shooter built in **4 day** with a **19-person team** for **GMTK Game Jam 2026** (10,600+ entries)
- Authored the game's **visual effects** layer in **GLSL**. Including enemy disintegration, screen-space vignette and desaturation, bloom, muzzle flashes, projectile trails
- Built enemy archetypes with a wave-based difficulty scaler and the full menu/HUD flow with animations

TECHNICAL SKILLS

Languages (Advanced): C#, Java

Languages (Intermediate): Python, C/C++, JavaScript/Typescript, CSS, Rust, GDScript

Frameworks: Unity, Raylib, NodeJS, React, Svelte, Vue, Tauri

Tools: Git, Github, Github Actions, Docker, Jira, Confluence, Trello